ASYNCHRONOUS PHYSICAL DATA LOGISTICS NETWORK

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Title: ASYNCHRONOUS PHYSICAL DATA LOGISTICS NETWORK

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Abstract

A system and method for a logistics network that routes large datasets as physical payloads, operating as a physical analogue to a packet-switched telecommunications network. The system, termed an Optically Attached Logistic Network (OALNet), comprises a plurality of interconnected hubs, a fleet of transport vehicles equipped with high-capacity internal storage, and a centralized routing and management system. Data is ingested from a client's portable cartridge at a source hub and transferred to a vehicle's onboard storage. The data payload is then physically transported, potentially being transferred between different vehicles at intermediate hubs, until it reaches the destination hub, where it is transferred to a local cartridge for last-mile delivery. This method of routing the data itself, rather than the original physical media, creates a scalable and efficient network for physical data transport.

Background of the Invention

While the internet provides a powerful network for routing digital packets of data, its architecture is not always optimal for the transfer of extremely large datasets, which can suffer

from bandwidth limitations and security vulnerabilities. The concept of physically moving data storage media has existed for decades, but it has always been an ad-hoc process without the sophisticated routing, sorting, and management infrastructure of a true network. As the volume of data continues to grow, there is a need for a formalized, scalable, and secure network system that is designed from the ground up to handle the logistics of physical data transport.

Summary of the Invention

The present invention is a system for an Asynchronous Physical Data Logistics Network (OALNet), which applies the principles of packet-switched networking to the physical world. The "packets" in this network are large datasets of ciphertext, and the "routers" are physical hubs where this data is transferred between high-capacity storage units.

The system comprises three main components: 1) A plurality of physical hubs, which act as high-speed data exchanges. 2) A fleet of transport vehicles (e.g., trucks, aircraft), each equipped with its own massive, high-bandwidth Optically-Attached Storage (OAS) system. 3) A central logistics management system that functions as the network's control plane, calculating optimal physical routes for the data payloads.

The process begins when a sender brings a portable data cartridge to a source hub. The data (ciphertext) is transferred from the cartridge to the onboard OAS of a long-haul transport vehicle. That vehicle transports the data to the next hub, where the data may be transferred to another vehicle's OAS, analogous to a packet moving to the next router. This continues until the data reaches the destination hub. At the destination hub, the data is transferred from the final transport vehicle to a local, "last-mile" portable cartridge for delivery to the recipient. This method of treating data as a fluid payload that is routed and transferred between network assets is a core innovation.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, de-

fense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the overall architecture of the Asynchronous Physical Data Logistics Network (OALNet).

FIG. 2 is a diagram illustrating the secure data handling protocol, showing the separate paths for the data cartridge and its decryption key.

Detailed Description of the Invention

Referring to **FIG. 1**, the OALNet system architecture is shown. The network consists of multiple physical hubs (101, 102, 103). A sender (106) brings a portable data cartridge to the source hub (101). At the hub, the data is transferred via a high-bandwidth optical link to the onboard Optically-Attached Storage (OAS) array of a transport vehicle (104a). The vehicle (104a) transports the data to an intermediate hub (102). At this hub, the data payload may be transferred to a different vehicle (104b) that is heading towards the final destination hub (103). This transfer between vehicles is analogous to packet routing in a digital network. Finally, at the destination hub (103), the data is transferred to a local cartridge for delivery to the recipient (107). A central Logistics Management System (105) orchestrates all data transfers and vehicle movements.

FIG. 2 illustrates the secure data protocol. A sender (201) has a plaintext dataset (202). The data is encrypted using an encryption key (203), resulting in ciphertext (204). The ciphertext (204) is loaded onto a portable data cartridge (205) and brought to a hub. The ciphertext payload is then transferred and routed through the OALNet's physical logistics channel (206) as described in FIG. 1. The encryption key (203), however, is sent to the recipient (207) via a completely separate, out-of-band channel (208), for example, a secure internet transmission. The recipient (207) uses the received key to decrypt the data delivered by the final cartridge. At no point during physical transit does the carrier have possession of both the data and the key required to read it.

Claims

What is claimed is:

1. A system for a physical data logistics network, comprising:

    - a plurality of transport vehicles, each vehicle equipped with a high-capacity, high-bandwidth internal data storage system;
    - a plurality of geographically distributed physical hubs, each hub configured to facilitate the high-speed transfer of data between said vehicles' internal storage systems or between a vehicle's internal storage system and a portable data cartridge; and
    - a logistics management system configured to calculate a route for a data payload, said route comprising at least one physical transport leg on a vehicle and at least one data transfer operation at a hub.

2. The system of claim 1, wherein the internal data storage systems and the portable data cartridges are configured for data transfer via a high-bandwidth parallel interface.

3. The system of claim 2, wherein the high-bandwidth parallel interface is a massively parallel optical interface.

4. A method for routing a data payload through a physical logistics network, the method comprising the steps of:

    - receiving a data payload from a source cartridge at a first hub;
    - transferring the data payload to a first internal storage system of a first transport vehicle;
    - transporting the data payload to a second hub via the first transport vehicle; and
    - at the second hub, transferring the data payload from the first internal storage system to a second storage system.

5. The method of claim 4, wherein the second storage system is an internal storage system of a second transport vehicle.

6. The method of claim 4, wherein the second storage system is a portable data cartridge for final delivery.

Drawings

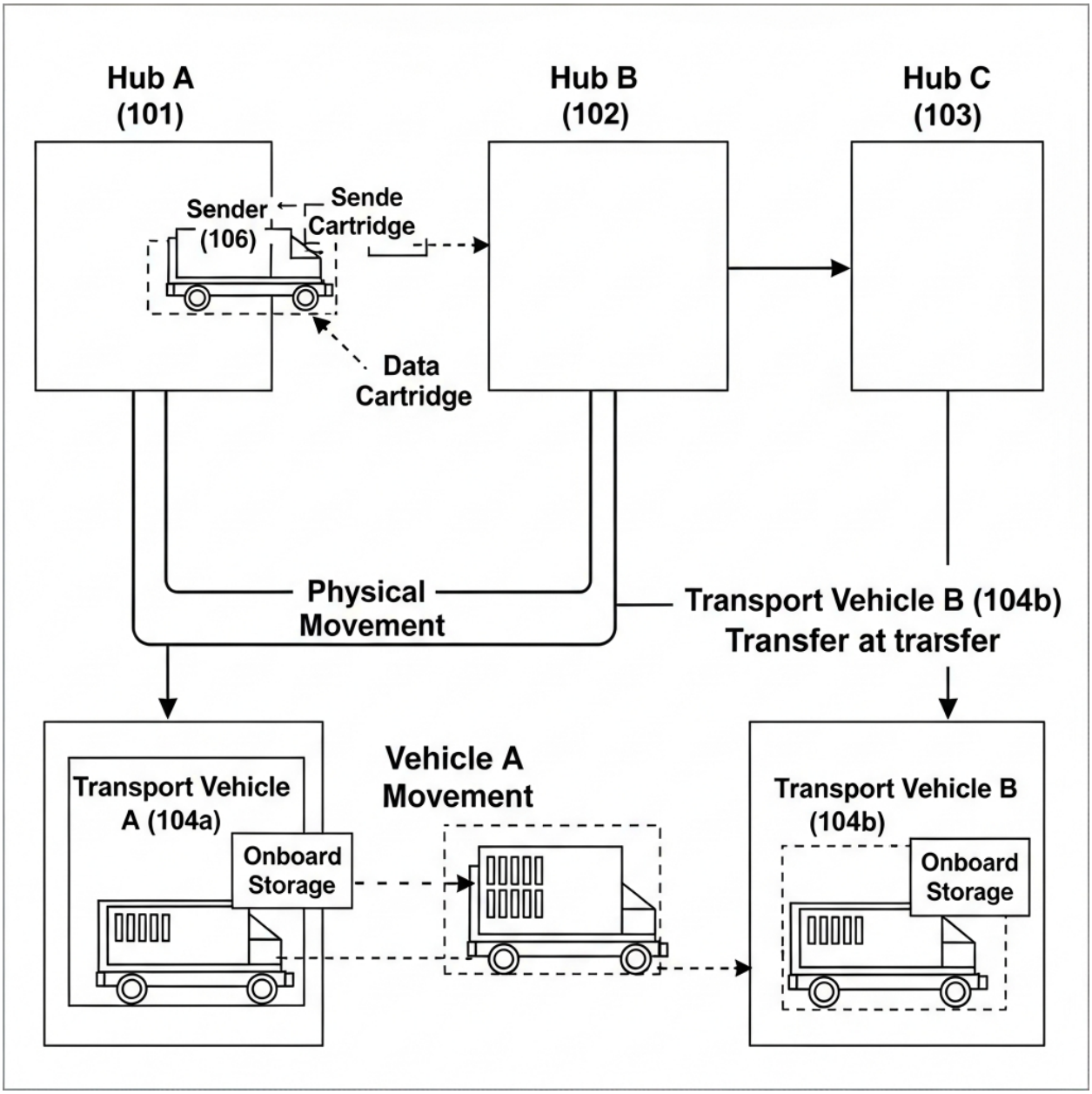


Figure 1: A block diagram illustrating the overall architecture of the Asynchronous Physical Data Logistics Network (OALNet).

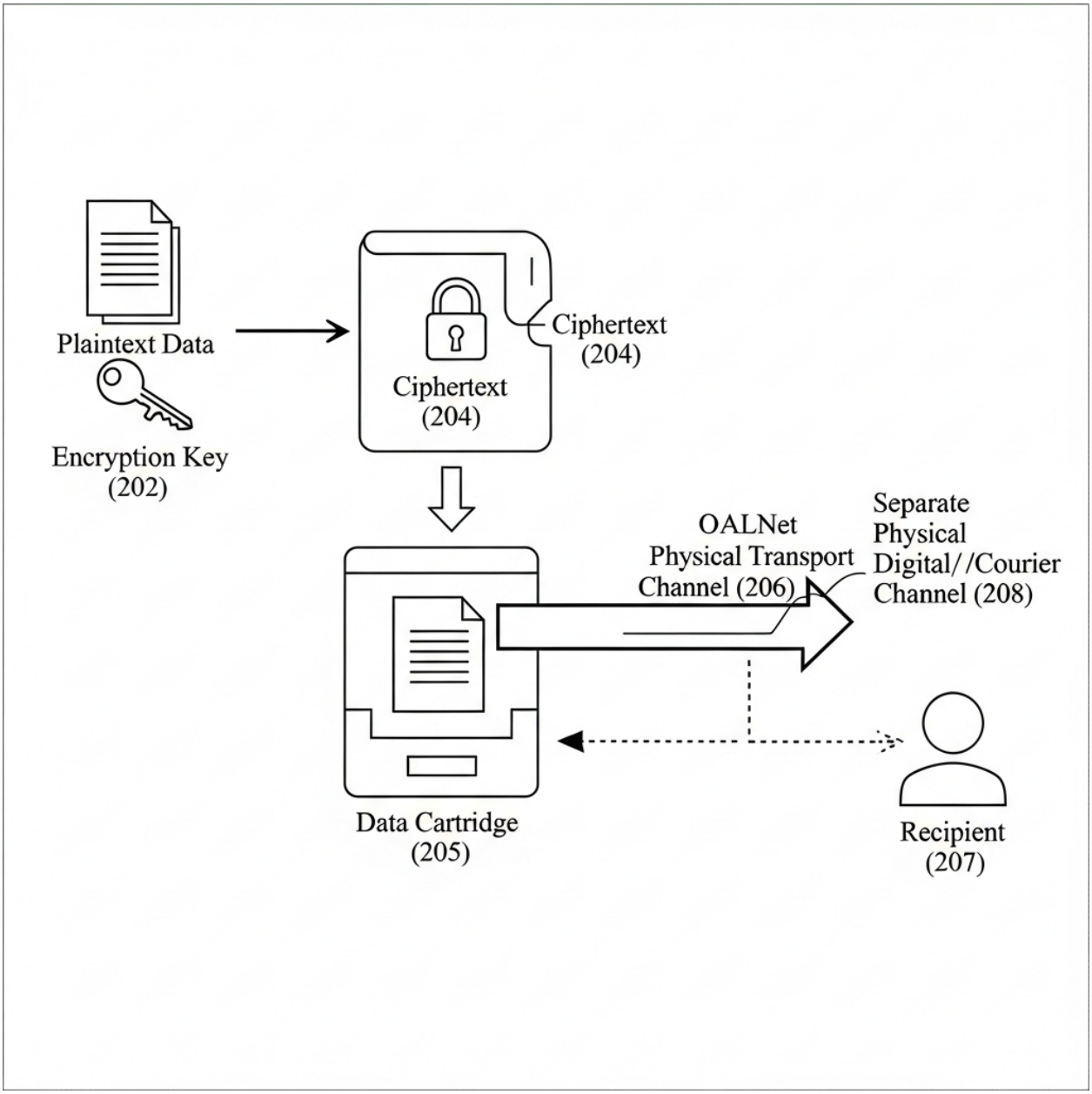


Figure 2: A diagram illustrating the secure data handling protocol, showing the separate paths for the data cartridge and its decryption key.